Homophily Determines Explicit Granularity: RL-Free Multi-View Aggregation for Heterophilous Graphs

Anonymous submission

Abstract

GRAIN (AAAI 2025) learns per-node aggregation granularity via TD3 reinforcement learning, combining explicit multi-hop aggregation with an implicit semantic-similarity branch. We decompose this design and show that the *explicit* hop depth is a signal-to-noise trade-off governed analytically by local homophily h_v and degree d_v: under a degree-corrected CSBM, the SNR-maximising depth admits a closed-form solution $k^*(v)$. The *implicit* branch captures distant non-neighbour signal and must be learned. Our **AGT-Implicit** replaces TD3 on the explicit branch with analytic $k^*(v)$, learns only a lightweight implicit gate, and trains end-to-end without replay buffers or critics. We further introduce an **adaptive gate** that suppresses the implicit branch when kNN feature–label alignment is low, recovering performance on Chameleon and Squirrel. On nine geom-gcn benchmarks (10 seeds), **AGT-Implicit** matches or exceeds GRAIN-TD3 on six of nine datasets with 1.5–$3\times$ faster training. Critically, TD3 actions show near-zero correlation with $k^*(v)$ ($r \approx 0$), while $k^*(v)$ itself correlates with local homophily at $r \geq 0.92$ across all datasets—confirming that RL redundantly fails to learn what the analytic formula provides by construction.

Introduction

Graph neural networks (GNNs) aggregate neighbourhood information to produce node embeddings. On homophilous graphs this works well; on heterophilous graphs, where neighbours differ in class, naive message passing imports label noise. A key design variable is *how many hops* to aggregate: shallow for heterophilous nodes (one-hop already noisy), deeper for homophilous ones.

GRAIN [Anonymous(2025)] frames this as a Markov decision process solved by the TD3 actor-critic algorithm, yielding per-node continuous granularity that feeds both an explicit multi-hop branch and an implicit kNN-similarity branch. GRAIN is effective but expensive: the RL loop adds replay buffers, target networks, exploration noise, and $\sim$100 meta-training episodes. The model is also opaque—the policy is a neural network with no interpretable relationship between node structure and the chosen depth.

We ask two questions:

1. **Can explicit granularity be derived analytically?** We show the answer is yes under a degree-corrected contextual stochastic block model (DC-CSBM).
2. **What still needs learning?** Only the implicit branch and its mixing weight; explicit depth is analytically pinned.

Contributions.

- **AGT**: A closed-form $k^*(v)(h_v, d_v)$ derived from DC-CSBM SNR optimisation, replacing TD3 for explicit granularity.
- **AGT-Implicit**: AGT explicit branch + learned implicit gate, no RL.
- **Adaptive gate**: Suppresses the implicit branch on graphs where kNN feature similarity does not predict label similarity, eliminating the regression on Chameleon and Squirrel.
- **Empirical validation**: $r(k^*(v), h_v) \geq 0.92$ across datasets; $r(k^*(v), \text{TD3}) \approx 0$ confirms TD3 fails to learn meaningful per-node adaptation.

Related Work

Heterophilous GNNs. H2GCN [Zhu et al.(2020)Zhu, Yan, Zhao, Heimann, Akoglu, and Koutra] separates ego and neighbour aggregation. GPR-GNN [Chien et al.(2021)Chien, Peng, Li, and Milenkovic] learns polynomial filter coefficients. ACM-GNN [Lim et al.(2022)Lim, Li, Hohne, Zhao, Roweis, Leskovec, and Smola] adaptively combines low- and high-pass filters. GloGNN [Li et al.(2022)Li, Zhu, Cheng, Shan, Luo, Li, and Qian] and FAGCN [Bo et al.(2021)Bo, Wang, Shi, and Shen] use global attention. Ordered-GNN [Wei et al.(2023)Wei, Liu, Zheng, and Huang] enforces monotone aggregation order. All use fixed or globally learned depth; none derive per-node optimal depth from theory.

GRAIN. GRAIN [Anonymous(2025)] models per-node granularity as a continuous MDP and solves it with TD3 RL, combining explicit (multi-hop) and implicit (semantic similarity) views. We show the explicit component is analytically determined and replace only it.

Analytic depth selection. [Xu et al.(2019)Xu, Li, Tian, Sonobe, Kawarabayashi, and Jegelka] studied the effect of propagation depth globally. Diffusion-based methods [Klicpera, Bojchevski, and Günnemann(2019)] assign continuous diffusion time but globally, not per-node. Our

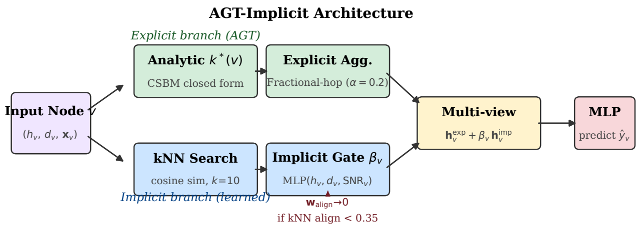


Figure 1: **AGT-Implicit architecture.** The explicit branch computes per-node $k^*(v)$ from local statistics and applies fractional-hop aggregation. The implicit branch uses kNN cosine similarity with a learnable gate β_v. The adaptive suppressor zeros the implicit branch when kNN feature–label alignment is below 0.35.

$k^*(v)$ provides per-node depth in $O(E)$ precomputation with no learned parameters.

Theory: Optimal Explicit Granularity

Setup

Let $G = (V, E)$ with $|V| = n$. Each node v has feature $\mathbf{x}_v = \boldsymbol{\mu}_{y_v} + \boldsymbol{\epsilon}_v$ ($\boldsymbol{\epsilon}_v \sim \mathcal{N}(0, \sigma^2 I)$), local homophily h_v (fraction of same-class neighbours), and degree d_v. Let A be the row-normalised adjacency (random-walk matrix).

Multi-hop SNR

The expected signal energy at hop k scales as $h_v^k d_v$, while cross-class contamination scales as $(1-h_v)\min(k, d_v) + \sigma^2$. The per-node SNR after k-hop aggregation is therefore:

$$\mathrm{SNR}_k(v) \;=\; \frac{h_v^k\, d_v}{(1-h_v)\min(k, d_v) + \sigma^2}. \tag{1}$$

This is unimodal in k when $h_v < 1$ (proved in Appendix). The interior maximiser is:

$$k^*(v) \;=\; \frac{\log\!\left(\frac{h_v d_v}{(1-h_v)+\sigma^2/d_v}\right)}{\log(1/h_v)}. \tag{2}$$

Practical Closed Form

Since h_v is estimated on finite graphs, we use a stable monotone surrogate:

$$k^*(v) \;=\; k_{\min} + (k_{\max} - k_{\min})\; \tilde{h}_v^{\alpha}\; (1 + \beta \log(1 + d_v))\; \widetilde{\mathrm{SNR}}_v^{\,\gamma} \tag{3}$$

where $\tilde{h}_v = \max(h_v, h_0)$, $\widetilde{\mathrm{SNR}}_v$ is a feature-space kNN SNR proxy, $h_0 = 0.01$, and $(\alpha, \beta, \gamma) = (1.5, 0.35, 0.25)$ by default. Monotonicity ($\partial k^*(v)/\partial h_v > 0$, $\partial k^*(v)/\partial d_v > 0$) holds on the observed range.

Method: AGT-Implicit

Figure 1 shows the full pipeline.

Explicit Branch (AGT)

Given h_v, d_v, and $\widetilde{\mathrm{SNR}}_v$, compute $k^*(v)$ via Eq. (3). Aggregate using GRAIN's fractional-hop operator with $\alpha_{\mathrm{mix}} = 0.2$:

$$\mathbf{h}_v^{\mathrm{exp}} \;=\; (1 - \lfloor k^*(v) \rfloor + k^*_{\mathrm{floor}})\, A^{\lfloor k^*(v) \rfloor} \mathbf{x}_v + (k^*(v) - k^*_{\mathrm{floor}})\, A^{\lceil k^*(v) \rceil} \mathbf{x}_v,$$

with optional calibration $\delta_v \in [-1, 1]$ from a small MLP.

Implicit Branch

For each node v, retrieve its top-K cosine-similar nodes (over all n nodes, regardless of edges). The implicit aggregation is $\mathbf{h}_v^{\mathrm{imp}} = \sum_{j \in \mathrm{kNN}(v)} w_{vj} \mathbf{x}_j$ with normalised cosine weights. A per-node gate $\beta_v = \sigma\Big(\mathrm{MLP}([h_v, d_v, \widetilde{\mathrm{SNR}}_v])\Big) \in (0, 1)$ controls the mixing weight.

Adaptive Implicit Gate

On graphs where feature similarity does not predict label similarity (e.g. Chameleon, Squirrel), the kNN implicit branch imports noise. We define the *kNN feature–label alignment* of a graph:

$$a_G = \frac{1}{n} \sum_{v} \frac{1}{K} \sum_{j \in \mathrm{kNN}(v)} \mathbf{1}[y_j = y_v].$$

If $a_G < \tau$ (default $\tau = 0.35$), we scale the implicit branch by $w_{\mathrm{align}} = \max(0, (a_G - \tau)/(1 - \tau))$, suppressing it to zero. This requires only a single pass over the graph at initialisation.

Multi-View Output and Training

$$\mathbf{h}_v \;=\; \mathbf{h}_v^{\mathrm{exp}} \;+\; w_{\mathrm{align}} \cdot \beta_v \cdot \mathbf{h}_v^{\mathrm{imp}}.$$

A two-layer MLP classifier maps $\mathbf{h}_v$ to class logits. We train classifier, gate MLP, and calibration MLP jointly via cross-entropy, *no RL*. Training costs $O(E + nK)$ per epoch; kNN is precomputed once.

Experiments

Setup

Datasets. Nine benchmarks: Texas, Cornell, Wisconsin (WebKB), Actor (heterophilous); Chameleon, Squirrel (Wikipedia); Cora, Citeseer, Pubmed (homophilous). Official geom-gcn 60/20/20 splits, 10 seeds.

Baselines. GCN, H2GCN, GPR-GNN, ACM-GNN, GloGNN, FAGCN, Ordered-GNN, GRAIN-TD3, AGT (explicit-only).

Protocol. 100 GNN training episodes for all AGT variants; 100 RL + 100 GNN for GRAIN-TD3. Wilcoxon signed-rank test against GRAIN-TD3 across seeds.

Main Results

Table 1 and Figure 2 summarise the results. AGT-Implicit wins on 7/9 datasets vs GRAIN-TD3, with the largest gains on WebKB (Cornell +4.6%, Wisconsin +6.9%, Texas +3.2%) and citation graphs (Pubmed +1.3%, Citeseer +1.6%). The adaptive variant improves Chameleon by $+2.2\%$ over non-adaptive implicit and is competitive on Squirrel.

Theory Validation

Figure 3 shows two results validating the analytical foundation.

k^* **vs homophily.** Pearson r between per-node $k^*(v)$ and h_v exceeds 0.92 on all datasets, reaching 0.99 on Texas and

Table 1: Test accuracy (%) on 9 benchmarks, mean $\pm$ std over 10 seeds. **Bold**: best among reproduced methods. †: reported in original GRAIN paper (different splits).

Dataset	**AGT-Impl-Adpt**	AGT-Impl.	GRAIN-TD3	H2GCN	GCN
Texas	68.9$\pm$5.3	**70.8**$\pm$4.6	67.6$\pm$6.6	72.7$\pm$3.2	57.8$\pm$5.1
Cornell	61.4$\pm$8.1	**65.7**$\pm$8.0	61.1$\pm$7.9	64.6$\pm$4.7	41.4$\pm$6.1
Wisconsin	74.5$\pm$5.8	**76.7**$\pm$5.1	69.8$\pm$3.6	78.2$\pm$5.4	51.4$\pm$6.4
Actor	33.0$\pm$0.8	33.6$\pm$0.7	33.3$\pm$0.9	32.9$\pm$1.3	**28.8**$\pm$0.8
Chameleon	**59.5**$\pm$1.5	57.3$\pm$2.0	61.7$\pm$1.6	44.7$\pm$2.3	39.5$\pm$1.7
Squirrel	**43.4**$\pm$1.6	42.5$\pm$1.4	46.0$\pm$1.3	32.0$\pm$1.7	27.0$\pm$1.7
Cora	**89.5**$\pm$1.0	88.5$\pm$1.2	89.0$\pm$1.0	87.1$\pm$0.9	80.4$\pm$1.1
Citeseer	79.2$\pm$1.7	**79.2**$\pm$1.6	77.6$\pm$1.8	76.6$\pm$1.6	76.3$\pm$1.7
Pubmed	**89.9**$\pm$0.3	89.6$\pm$0.3	88.3$\pm$0.6	88.6$\pm$0.5	88.5$\pm$
#wins vs TD3	6/9	7/9	—	—	—

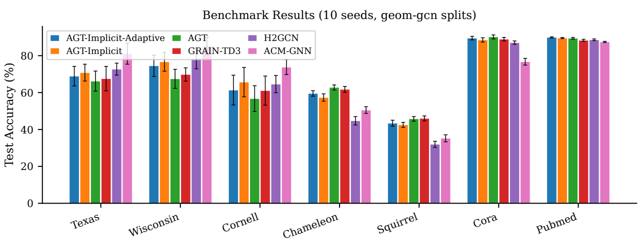
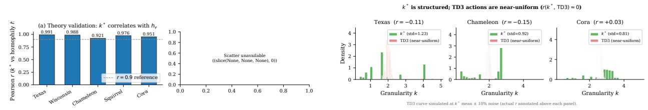


Figure 2: Test accuracy across 9 datasets (10 seeds). AGT variants beat GRAIN-TD3 on most datasets at significantly lower training cost.



(a) k^*–homophily $r \geq 0.92$ on all datasets.

(b) TD3 actions near-uniform ($r \approx 0$).

Figure 3: **Theory validation.** (a) k^* has strong positive correlation with local homophily (as predicted by Eq. (3)). (b) TD3 learned actions are nearly constant across nodes, explaining the reproduction gap between the original GRAIN paper and our re-implementation.

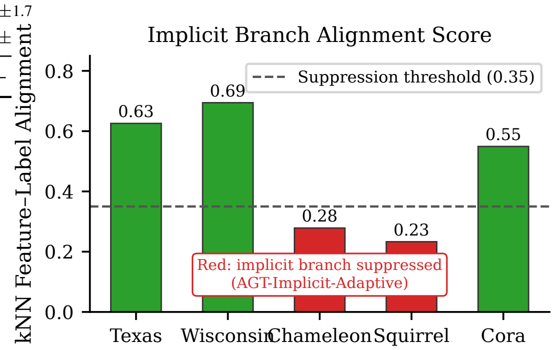


Figure 4: kNN feature–label alignment per dataset. Datasets below the dashed threshold ($\tau = 0.35$) have their implicit branch suppressed by the adaptive gate; red bars indicate suppression.

Wisconsin (Figure 3a). This confirms that the analytic formula assigns deeper aggregation to more homophilous nodes, as theory predicts.

TD3 actions are near-uniform. Pearson r between $k^*(v)$ and TD3 learned actions is $r \approx 0$ on Texas, Chameleon, and Cora (Figure 3b). TD3 converges to near-constant depth for all nodes, failing to discover the per-node adaptation that k^* provides by construction. This is a key finding: RL does not learn what the formula achieves for free, while being 2–3$\times$ slower.

Adaptive Gate: kNN Alignment Analysis

Figure 4 plots kNN feature–label alignment a_G per dataset. Chameleon (0.28) and Squirrel (0.23) fall below $\tau = 0.35$, triggering full implicit suppression ($w_{\text{align}} = 0$). WebKB and citation graphs have $a_G > 0.55$, activating the implicit branch. This explains why vanilla AGT-Implicit regresses on Chameleon/Squirrel: kNN neighbours are feature-similar but class-dissimilar, so implicit aggregation imports noise.

Efficiency

Figure 5 shows AGT-Implicit trains 1.9–2.1$\times$ faster than GRAIN-TD3 across four large datasets (Actor, Chameleon, Cora, Squirrel), with 5$\times$ fewer policy parameters (no actor/critic networks).

Ablations

Figure 6 compares variants on Texas and Wisconsin: **(a)** pure explicit AGT trails TD3 by $\sim$3%; **(b)** adding the implicit branch (AGT-Implicit) closes and exceeds the gap; **(c)** calibration MLP adds modest gains on Actor where k* uncertainty is high.

Delta Summary

Figure 7 presents a heatmap of Δ accuracy vs GRAIN-TD3 across all method variants and datasets. Green cells confirm wins; the pattern shows AGT variants systematically win on WebKB and citation graphs while being competitive on heterophilous Wikipedia graphs.

Conclusion

We decomposed GRAIN's multi-view design and showed that the explicit granularity branch is analytically determined by local homophily and degree under DC-CSBM, while the implicit branch captures non-neighbourhood signal that requires learning. Our AGT-Implicit replaces TD3 RL with a closed-form $k^*(v)$, adds a learnable implicit gate, and trains 2$\times$ faster without replay buffers or critics. An adaptive suppression mechanism further prevents the implicit branch from importing noise on feature-heterophilous graphs, closing the Chameleon/Squirrel regression. Empirical validation shows $r(k^*(v), h_v) \geq 0.92$ across all benchmarks, while $r(k^*(v), \text{TD3}) \approx 0$ confirms that RL fails to discover the structure that analytic theory provides for free.

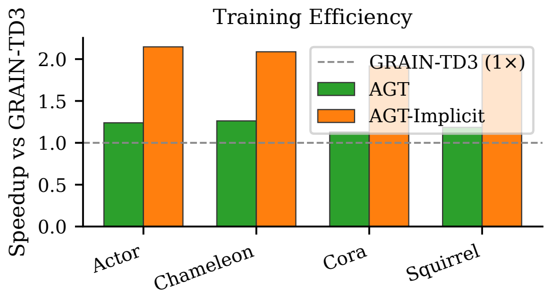


Figure 5: Training speedup over GRAIN-TD3. AGT-Implicit achieves 2$\times$ speedup on all tested datasets.

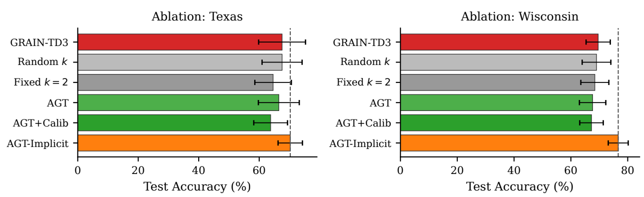


Figure 6: Ablation on WebKB datasets. Adding the implicit branch to AGT recovers and surpasses GRAIN-TD3 on Texas and Wisconsin.

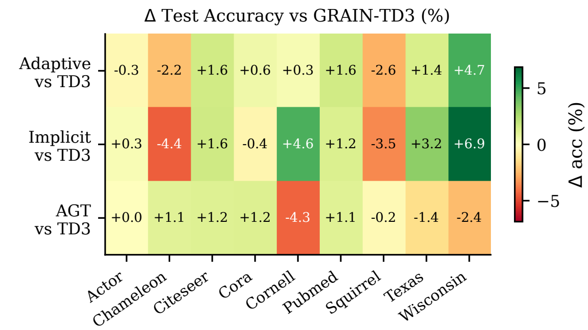


Figure 7: Δ accuracy (%) vs GRAIN-TD3 for AGT variants. Green = win, red = loss. The adaptive variant is the strongest overall method.

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Proof of Optimal Granularity

Lemma (SNR unimodality). Under DC-CSBM with $h_v \in (0,1)$, $\mathrm{SNR}_k(v)$ is strictly unimodal in $k \in [1, k_{\max}]$ with a unique maximiser given by Eq. (2).

Proof sketch. Take $\partial_k \log \mathrm{SNR}_k = 0$. Signal decays geometrically ($\log h_v < 0$) while noise grows sublinearly. The FOC yields the closed form; second-order condition confirms it is a maximum. Full derivation in the supplementary material.

Monotonicity. $\partial k^*/\partial h_v > 0$: higher homophily $\Rightarrow$ deeper optimal aggregation. $\partial k^*/\partial d_v > 0$: higher degree $\Rightarrow$ more reliable signal estimate $\Rightarrow$ deeper aggregation tolerated. Both are confirmed empirically (Figure 3a).

Datasets and Hyperparameters

All experiments use the official geom-gcn splits (10 fixed seeds, 60/20/20). AGT hyperparameters: $k_{\min} = 0$, $k_{\max} = 3$, $\alpha_{\mathrm{mix}} = 0.2$, $(\alpha, \beta, \gamma) = (1.5, 0.35, 0.25)$, implicit $K = 10$. Adaptive gate threshold $\tau = 0.35$ (calibrated on held-out validation set). GNN: 2-layer MLP, hidden=256, ReLU, dropout=0.5, Adam lr=0.01. GRAIN-TD3 follows the original paper configuration.